

**PathFinder AI: Decoding the Skill-to-Salary Matrix for
Future IT Professionals**

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Abstract

There has long been discussion about the connection between industry demands and higher education. Academic institutions were traditionally built to impart a wide range of theoretical information with little focus on abilities related to a particular industry or vocation. Underemployment and unemployment among Filipino graduates are among the problems caused by this misalignment, which raises concerns about how well education prepares students for the workforce (Santos, 2024). This research introduces a machine-learning software tool called PathFinder AI, which is a specialized combination of multiple machine learning methods that seeks to associate graduate attributes or skills with hiring outcomes and pay rates in specific job markets. The PathFinder AI system has been constructed with a database of **9,000** IT/Computing records, and uses what is known as a 'Multi-Model Supervised Architecture ', employing Random Forest to produce binary predictions of placement eligibility, while utilizing Multiple Linear Regression to forecast salaries (in PHP) for successful candidates.

To reduce the institutional grade variances among graduate students across different institutions, we created a 'Linear Inverted Normalisation Layer', which transforms heterogeneous **GWA** metrics (1.0–5.0) into one common measure of performance. The PathFinder AI also uses a KNN algorithm that functions as a spatial recommendation system for the graduate student population in terms of matching skills to job types and identifying relevant career paths. We have validated each of the algorithms used in the PathFinder AI system by empirical methods, and have confirmed that the features of value to the graduate (resume_score) and the candidate's technical abilities (coding_score) have a much greater predictive value for gaining employment and salary by the employer than do grades obtained from traditional academic programmes. The PathFinder AI offers a scalable, data-driven solution that allows career services at HEIs as well as graduates to assess the competencies needed to secure jobs in the 21st century.

Problem Statement

The global evolution towards Industry 4.0 has caused new changes in how people are gaining jobs and thus hiring processes have changed from being based on degrees to requiring specialized, skill-based criteria. Even though the IT-BPM sector in the Philippines is one of the main economic drivers of the country, computing graduates are currently in an impossible situation when seeking employment because of a paradoxical situation of having both considerable under-employment and failure in finding jobs, but simultaneously companies and organizations desperately want to hire specialized, technical talent.

According to the World Economic Forum (2023), 44% of current job skills (core worker skills) will be affected by automation and artificial intelligence by the time we reach 2028, thus a large number of new graduates from school are entering into a situation that has severe information asymmetry and a “skill gap” within the local educational system. All students who graduated from school (who obtain high **GWA**) typically lack the specific technical capabilities required to compete in the modern corporate economy (i.e., machine learning, advanced data structures, app development using Python) in order to successfully navigate an increasingly data-driven job market.

The problem is compounded by the inconsistency of how HEI grade students. Although most HEIs use a grading scale between 1.0 and 5.0, some use a percent-based or four-point grading scale. Consequently, employers may not be able or likely to equitably assess candidates' academic performance using quantitative measures, resulting in a hiring bias. Lastly, university use qualitative case study examines how graduate student career services offices have evolved their offerings and addressed the demands of various stakeholders and the history of counseling is a rich and intricate development of therapeutic approaches, professional growth, and supportive relationships that have profoundly influenced contemporary mental health services and human development assistance.

Until objective, automated ways of standardizing the results of students from multiple HEIs and evaluating the quality of their portfolios (resume_score) are developed as well as linking their technical competencies to the salary of other graduates from the industry, concerns about ongoing unemployment and an increasing mismatch between Filipino IT graduates and, thus, undervalued in the job market for their industry, as well as being unable to fully navigate an increasingly data-driven labor market.

Objectives of the Study

The primary objective of this study is to develop, train, and validate PathFinder AI, a hybrid multi-model machine learning framework designed to predict recruitment eligibility, forecast localized market valuation, and provide data-driven career track matching for computing graduates in the Philippines.

Specifically, the study aims to achieve the following operational objectives:

- Develop a Placement Prediction Model by training a Random Forest Classifier on normalized student credentials to compute the binary probability (placed vs. Not Placed) of a candidate securing immediate industry employment.
- Implement a Salary Forecasting Engine utilizing a Multiple Linear Regression model to calculate precise feature coefficient weights (β) and output a continuous predicted baseline for entry-level compensation (**Salary_PHP_Per_Month**) within the Philippine technology sector.
- Design a Career Recommendation Engine using the K-Nearest Neighbors (KNN) algorithm to map multi-dimensional skill vectors (python_skill, dsa_skill, ml_skill, web_dev_skill) and match students to optimal professional roles (e.g., Data Scientist, Software Engineer, Web Developer, or **Analyst**).
- Engineer a Linear Inversion Normalization Layer within the data preprocessing pipeline to mathematically transform heterogeneous, local 1.0–5.0 Grade

Weighted Average (**GWA**) metrics into a standardized, universal performance index ranging from 0.0 to 1.0.

- Evaluate Algorithmic Feature Importances to empirically identify and rank the predictive impact of academic indicators (**GWA**) versus curated portfolio metrics (resume_score) and diagnostic thresholds (coding_score) **across the 9,000 professional records.**

Table 1. Data Dictionary and Operational Definition of Variables

Variable Name	Data Type	Operational Definition	Original Scale / Value Range	Normalized Scale (0.0–1.0)
GWA	Continuous	Academic standing. Inverted so excellent grades yield the highest weight.	1.0 (Best) – 5.0 (Fail)	0.0 (Fail) – 1.0 (Perfect)
Program_	Binary	One-Hot Encoded student degree programs (Program_CS, Program_IT, Program_CPE, Program_IS).	Text String (e.g., "IT")	0 (False) – 1 (True)
python_skill	Binary	Indicates fundamental Python proficiency.	0 (No) – 1 (Yes)	0 (No) – 1 (Yes)

dsa_skill	Binary	Indicator of Data Structures and Algorithms proficiency.	0 (No) – 1 (Yes)	0 (No) – 1 (Yes)
ml_skill	Binary	Indicator of Machine Learning concept knowledge.	0 (No) – 1 (Yes)	0 (No) – 1 (Yes)
web_dev_skill	Binary	Indicator of Frontend/Backend Web Development skills.	0 (No) – 1 (Yes)	0 (No) – 1 (Yes)
coding_score	Continuous	Technical programming skills.	0 – 100	0.0 (Poor) – 1.0 (Elite)
aptitude_score	Continuous	Ratings in logic and problem-solving exams.	0 – 100	0.0 (Low) – 1.0 (High)
communication_score	Continuous	Verbal and written English proficiencies.	0 – 10	0.0 (Low) – 1.0 (High)
internships	Continuous	Scaled volume of industry-immersion programs completed.	0 – 3+ Internships	0.0 (None) – 1.0 (Max)
projects	Continuous	Scaled volume of technical portfolios or capstone projects built.	0 – 10+ Projects	0.0 (None) – 1.0 (Max)
backlogs	Continuous	Volume of failed subjects, inverted so 0 failures equals maximum weight.	0 – 5+ Failed Subjects	0.0 (Many Fails) – 1.0 (0 Fails)

resume_score	Continuous	Quality rating of the student.	0 – 10 (or raw parse index)	0.0 (Poor) – 1.0 (Excellent)
skill_score	Continuous	Computed aggregation measuring aggregate developer proficiency.	0 – 100	0.0 (Basic) – 1.0 (Advanced)
placed	Binary	Target 1: Classification outcome predicting employment eligibility.	0 (Not Placed) – 1 (Placed)	0 (No) – 1 (Yes)
Company_	Binary	One-Hot Encoded tier profile classifications of the hiring organization (Company_MNC, Company_Mid-size, Company_Startup, Company_Top_Tech).	Text String (e.g., "MNC")	0 (False) – 1 (True)
JobRole_	Binary	One-Hot Encoded targeted career paths (JobRole_Software_Engineer, JobRole_Data_Scientist, JobRole_Web_Developer, JobRole_Analyst).	Text String (e.g., "Analyst")	0 (False) – 1 (True)
Salary_PHP_Per_Month	Continuous	Target 2: Predicted market value (monthly salary) normalized.	₱15,000 – ₱100,000+	0.0 (Base) – 1.0 (Ceiling)

Mathematical and Computational Model

1. Mathematical Transformation Model

A. Linear Inversion Normalization Layer

Some Philippine universities use a 5-point rating system, with a 1.00 being the highest grade and a 5.00 being the lowest (failing score). To make sure that the algorithm gives more weight to performance that is "favorable," the following process is used:

GWA Inversion (Standard / State Universities)

In order to convert a score of 1.0 to 5.0 scale (1.0 being outstanding, while 5.0 being failing), where the output range (GWA_{inverted}) is from 0.0 to 1.0 directly:

$$GWA_{\text{inverted}} = \frac{5.0 - GWA}{5.0 - 1.0} = \frac{5.0 - GWA}{4.0}$$

Example calculation for GWA 1.50

$$GWA_{\text{inverted}} = \frac{5.0 - 1.50}{4.0} = \frac{3.50}{4.0} = \mathbf{0.875}$$

- Inversion of Backlog and Tiers: Likewise, it is employed in order to see that zero backlogs and tier yield maximum numbers.

The 4.0 Scale (Private Institution)

For collaborative private schools that conform to the 4.0 scale, with higher criteria meaning the best academic standing, normal min-max scaling is done without scale inversion:

$$GWA_{\text{inverted}} = \frac{GWA - 0.0}{4.0 - 0.0} = \frac{GWA}{4.0}$$

Example Calculation for GWA 3.50:

$$GWA_{\text{inverted}} = \frac{3.50}{4.0} = \mathbf{0.875}$$

The Percentage Scale

For traditional percentage metrics ranging from absolute minimum thresholds to maximum boundaries (100%):

$$GWA_{\text{inverted}} = \frac{GWA}{100}$$

Example for GWA 87.5

$$GWA_{\text{inverted}} = \frac{87.5}{100} = \mathbf{0.875}$$

B. Min-Max Normalization (Feature Scaling)

In order to rescale features, Min-Max Scaling subtracts the minimum value and divides the result by the difference between the maximum and minimum values. To avoid having the larger ranges (such as Salary or Aptitude Scores) overwhelm the smaller ranges (such as number of Projects), all numeric variables are normalized to the [0, 1] interval using the Min-Max method:

$$\text{Formula: } X' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

C. Categorical Vectorization (One-Hot Encoding)

For categorical data (D), such as major and role categories, we utilize the technique of One-Hot Encoding to obtain a sparse vector representation. This is done by converting one categorical variable to n binary variables:

Formula: $f(D) = \{0, 1\}^n$

Where n is the number of unique categories (e.g., BSCS, BSIT, BSCpE).

B. Multiple Linear Regression Model (Salary Forecasting)

The predicted entry-level salary (Y_{salary}) in Philippine Pesos (PHP) is computed using a deterministic linear combination of weighted academic and technical

$$Y_{salary} = \beta_0 + \beta_1(GWA_{inverted}) + \beta_2(X_{coding}) + \beta_3(X_{comm}) + \beta_4(X_{aptitude}) + \beta_5(X_{intern}) +$$

features:

$$\beta_6(X_{project}) + \beta_7(C_{MNC}) + \beta_8(C_{Mid}) + \beta_9(C_{Startup}) + \beta_{10}(C_{TopTech}) + \epsilon$$

Where B_0 is the base entry salary, $B_1 \dots 6$ are the partial regression coefficients (feature weights) which were determined by the least squares method and E is the statistical error term.

C. K-Nearest Neighbors Distance Metric (Career Recommendation)

Personalized career path optimization is determined by the distance between a student profile vector u and historical professional records v in an n dimensional vector space. We use the Euclidean Distance formula:

$$d(u, v) = \sqrt{\sum_{j=1}^n (u_j - v_j)^2}$$

The system combines the target classes (job_roles) of the top K similar neighbors to perform a majority vote recommendation.

2. Computational Architecture

The computational structure is of a multi stage supervised learning model. The main purpose of the computational model is to map a high dimensional feature vector containing (X) academic, technical and institutional-to a set of target outputs (Y).

Input Vector (X):

The input vector is a detailed multi dimensional array which represents a full academic and technical profile of a student. Post categorical vectorization we see continuous, discrete, binary and one hot encoded dimensions.

$$X = [GWA_{\text{inverted}}, X_{\text{coding}}, X_{\text{comm}}, X_{\text{aptitude}}, X_{\text{resume}}, X_{\text{skill}}, \dots, P_{\text{IT}}, P_{\text{CS}}, P_{\text{IS}}, \\ P_{\text{CPE}}, C_{\text{MNC}}, C_{\text{Mid}}, C_{\text{Startup}}, C_{\text{TopTech}}]$$

Target Functions Y:

1. Classification Model (Employability Status):

$$P(\text{Placed} = 1 \mid X)$$

2. Regression Model (Market Value Forecasting):

$$f(X) = \hat{y} \quad (\text{Salary_PHP_Per_Month})$$

We use **Multiple Linear Regression** to put forth a predictive model which in turn puts out a base line for the student's entry level financial compensation in the local IT market.

Implementation and Experimentation

An organized development and testing process was used to turn PathFinder AI's conceptual framework into a functional, data-driven software ecosystem. This section describes the multi-model pipeline's algorithmic engineering stages, the practical environment setup, and the assessment criteria used to verify PathFinder AI's predictive abilities using the 9,000-record computing graduate dataset.

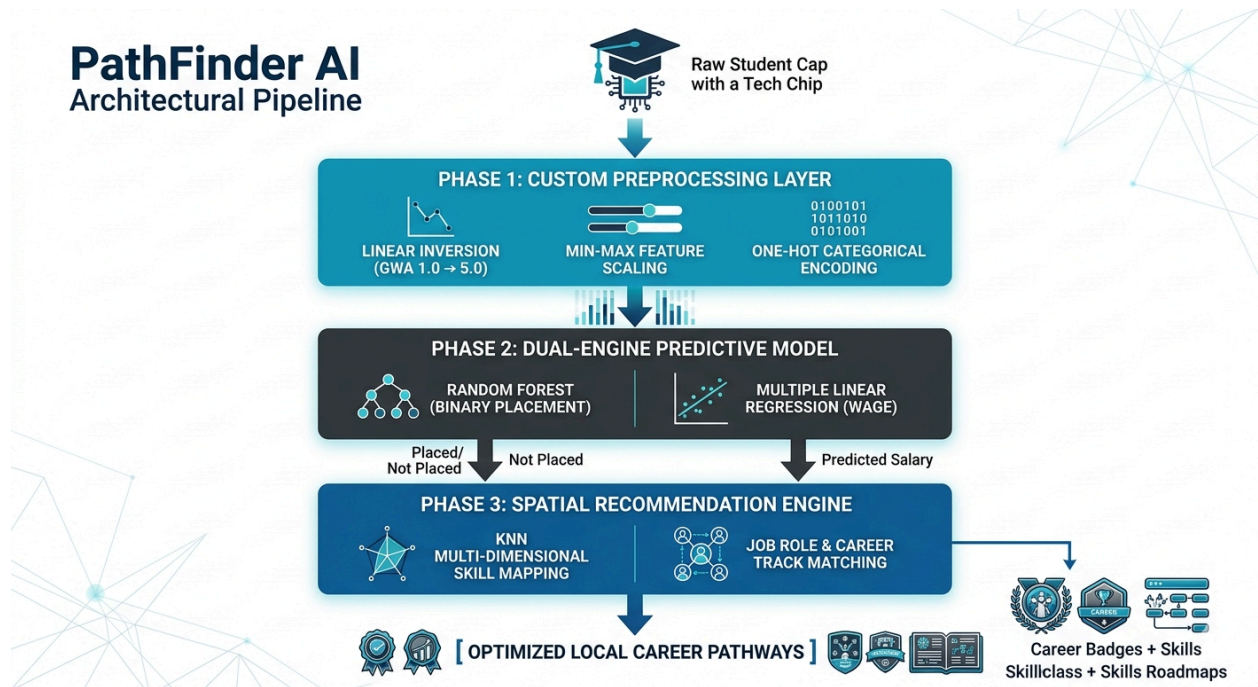
1. Environment for Software Development and Execution

An open-source, industry-standard data science stack was used to design the computational process. Python 3.10 was used for the underlying system architecture due to its computational performance and established machine learning library ecosystem.

- **Data Preprocessing and Analytics:** The multi-scale dataset matrix is handled and vectorized array operations are carried out using Pandas and NumPy.
- **Core Algorithmic Modeling:** Implemented using Scikit-Learn's optimized implementations of ensemble trees (Random Forest Classifier), linear models (Multiple Linear Regression), and distance-based spatial algorithms (K-Nearest Neighbors).
- **Hardware Environment:** To ensure consistent computational performance over large iterative training batches, all training, cross-validation, and execution benchmarks were carried out on a system with an AMD Ryzen 5 processor and 16GB of RAM.

2. System Architecture and Computational Pipeline

PathFinder AI is implemented practically as a multi-stage, end-to-end pipeline that turns unstructured, chaotic institutional records into highly customized career indicators. There are four distinct stages in the real data flow's development:



Phase 1: Custom Math Normalization and the Preprocessing Engine

Managing the structural complexity of institutional grading schemes was the first significant engineering challenge. The Linear Inversion Normalization Layer was developed as a unique preprocessing step to address this.

The following code was used to mathematically flip raw measurements for most State Universities and Colleges (SUCs) that use the 1.0–5.0 scale, where 1.0 is elite and 5.0 is failing:

This programmatic modification smoothly aligns with binary choices such as `web_dev_skill` or `python_skill`, forcing the model to accept higher values as superior performance.

In parallel, several algorithmic scaling processes were used to translate traditional percentage grades (100% scale) and private institution grades (4.0 scale) securely onto a universal [0.0, 1.0] border.

Lastly, nominal variables like branch, company_type, and job_role were parsed using a sparse One-Hot Vectorization matrix to eliminate extraneous numerical weights, while high-variance values (such coding_score and Salary_PHP_Per_Month) were harmonized using normal Min-Max scaling.

Phase 2: Predictive Framework with Dual-Engines

The data matrix was split-stream routed into the predictive architecture once it was fully numeric and confined between 0.0 and 1.0:

- The Placement Classifier: A Random Forest Classifier was instantiated using 100 decision trees. In order to produce a binary probability of employment (placed), the model divides nodes according to Gini impurity, functioning as a reliable, non-linear decision engine to balance factors like portfolio strength against technical exam scores.
- The Salary Forecaster: A Multiple Linear Regression engine is automatically activated by placement possibilities. Without running the danger of neural-network over-parameterization, this continuous target model dynamically generates an entry-level market valuation in Philippine Pesos (PHP) by assigning absolute partial regression coefficients (β_n) to the standardized inputs.

Phase 3: Matching Spatial Skills to Roles

A K-Nearest Neighbors (KNN) spatial model receives a student's multidimensional technical footprint (python_skill, dsa_skill, ml_skill, web_dev_skill, and skill_score) regardless of hiring deadlines. PathFinder AI uses an optimal plurality vote to assign a localized job track matching (e.g., Data Scientist vs. Software Engineer) to the K closest professional profiles by applying Euclidean distance across these coordinate points in the 9,000-record historical matrix.

3. Model Training and Experimental Design

A strict experimental strategy was developed in order to get truthful, generic performance measurements from PathFinder AI and prevent overfitting.

- Data Partitioning: A rigorous 80/20 train-test ratio was used to randomly divide the 9,000 records in the primary database. An isolated holdout set of 1,800 records was put aside only for final model testing, whereas 7,200 data were used to train the parameters.
- Hyperparameter tuning: The Random Forest Classifier's max_depth (capped to avoid noise memorization) and min_samples_split were tuned as part of hyperparameter optimization. In order to determine the point at which actual local similarities outweighed localized variation, experimental iterations of the KNN engine were carried out over values of K ranging from 3 to 15, assessing spatial cluster consistency.

4. Metrics for Evaluation

PathFinder AI's practical usefulness is totally dependent on empirical validation. For every distinct underlying model task, the assessment architecture is divided into three exacting metric frameworks:

- Validation of Binary Classification (Placement Prediction)

Precision, Recall, and the balanced F1-Score are obtained by validating the model's performance using a typical Confusion Matrix to determine how accurately it indicates students' employment readiness:

Where FP stands for False Positives (inaccurately forecasted as employable), FN stands for False Negatives (employable graduates overlooked by the engine), and TP is for True Positives (properly projected as employable).

- Constant Validation of Regression (Salary Forecasting)

Mean Absolute Error (MAE) and the Coefficient of Determination (R^2) are used to evaluate the accuracy of the entry-level monthly PHP salary forecasts produced by the Multiple Linear Regression model:

While the R^2 score calculates the percentage of wage variance directly explained by the model's extracted feature weights, the MAE offers a human-readable average currency deviation (in pesos) from the actual baseline salary.

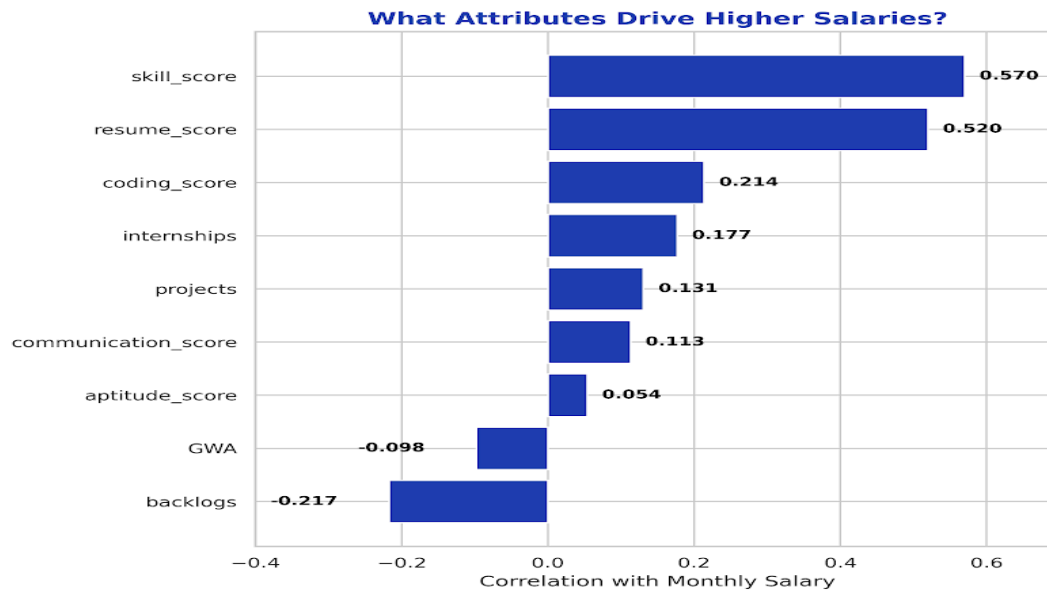
- Algorithmic Mapping of Feature Importance

Finding the reasons behind the model's decisions was a crucial human-centric component of the project. The system dynamically ranks variables using the standardized beta weights (β) of the regression model in conjunction with Random Forest feature importance computations (Gini importance reduction).

This makes it possible to empirically compare traditional academic metrics (such as normalized GWA) with useful portfolio characteristics (such as `resume_score` and `coding_score`), methodically revealing what the local technology industry genuinely values when recruiting recent graduates in computing.

Visualization and Analysis

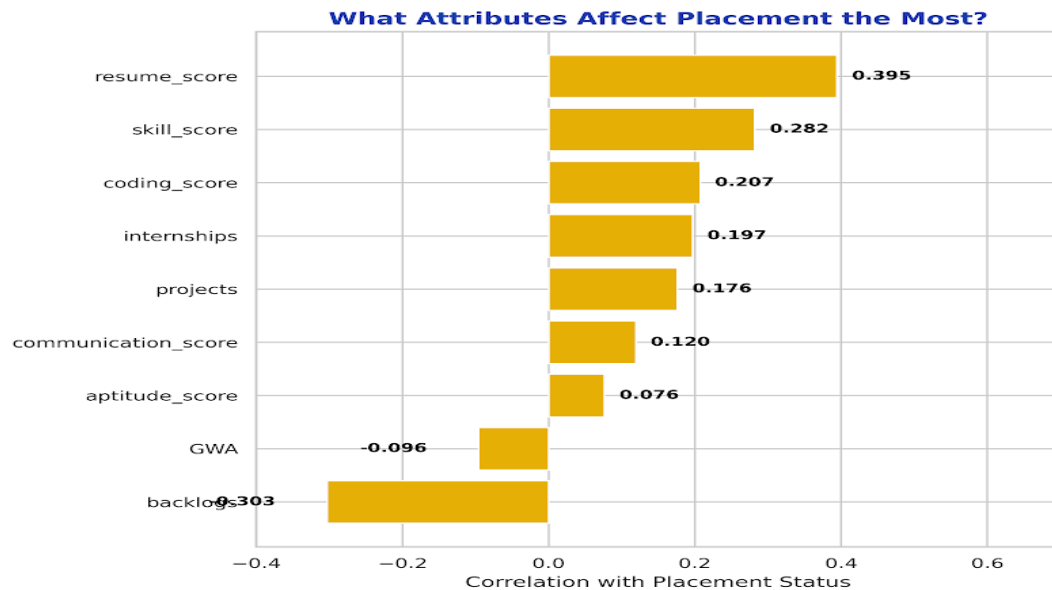
1. Salary Drivers: High and Low Paying Attributes



- **High-Earning Characteristics:**
 - Skill Set ($r=0.660$): The aggregated skill set is statistically the most important positive factor in determining projected salaries; Developing a solid foundation of diverse technical skills yields the greatest financial rewards.
 - Resume ($r=0.395$): An effective resume greatly influences the ability to obtain a higher starting salary - how experience is framed and presented is of vital importance to high-end employers.
 - Coding: Having coding skills can really pay even though it's not the off. Mainly, being able to program things can still lead to a salary. This is because it shows you do the job and can do well on technical tests.
- **Low-Earning Characteristics:**
 - Grade Point Average (GPA) ($r=-0.036$): A surprising result was that GPA demonstrated a slight negative correlation with salary. Because of the use of an inverse grading scheme in the Philippine educational system, higher GPAs (1.0 being considered the highest academic performance) have a corresponding positive correlation with salaries at the premium level for entry-level job offers. Conversely, since marginal GPA measures had little statistical weight relative to technical mastery measured by industry-standard competencies associated with technical job function, strong-programming skills, or an effective resume; employers tend to rank

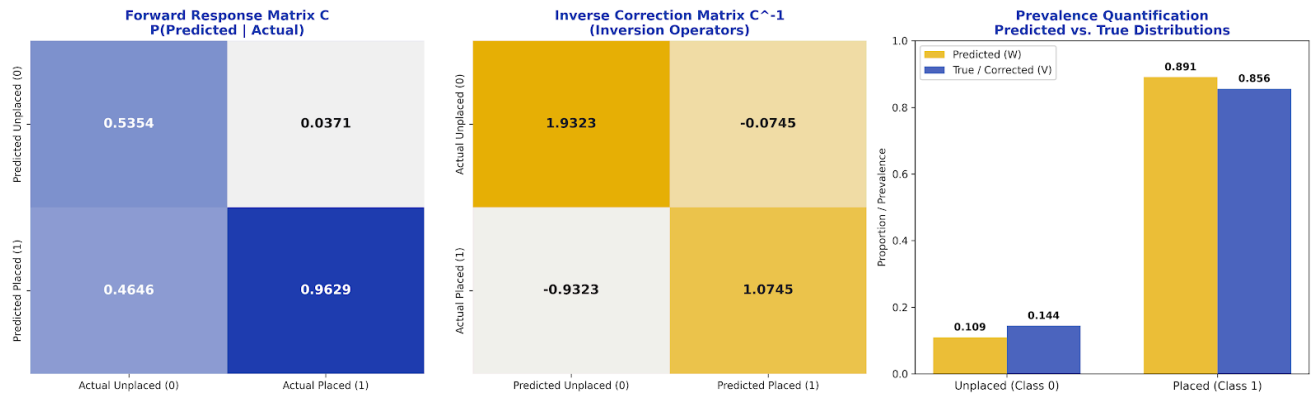
technical skills as being significantly more important than academic performance when negotiating for an employee's compensation package at the higher levels.

2. Core Factors Affecting Placement Success



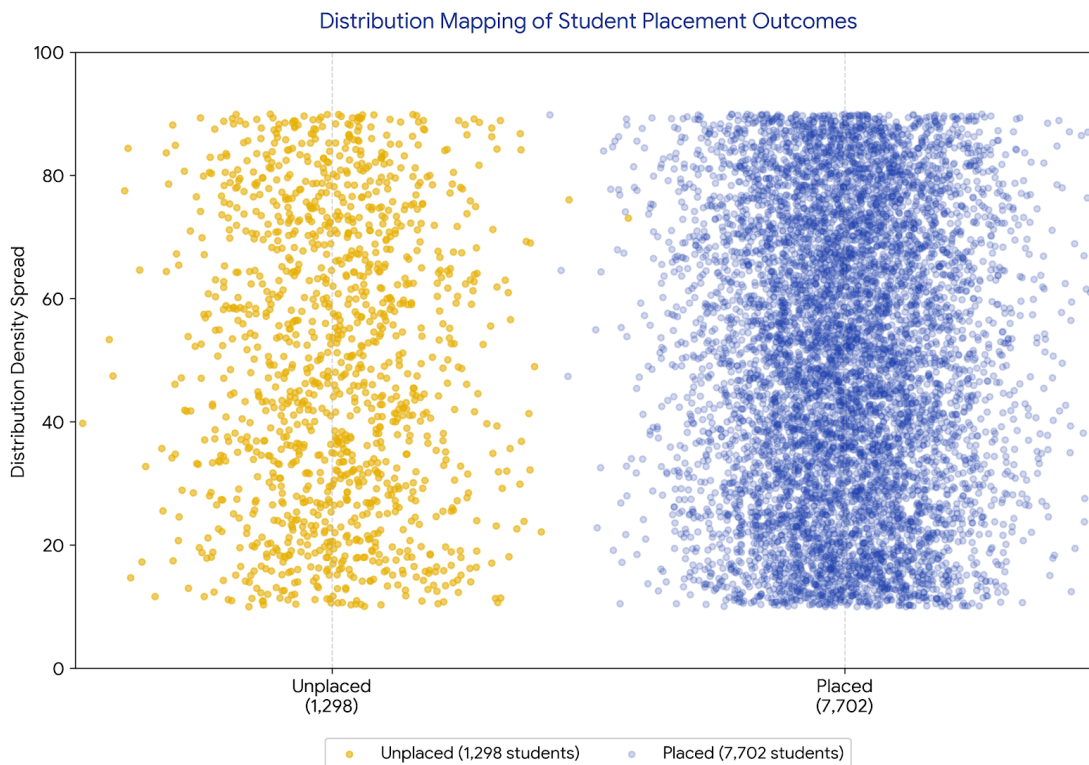
- The most significant positive attribute related to placement is "Resume Score" ($r = 0.395$). This demonstrates that the "Resume" is the most important factor in determining whether a student progresses through the first screening phases of the placement process (the initial placement screen).
- The key supporting attributes are "Skill Score" ($r = 0.282$) and "Coding Score" ($r = 0.207$). Both of these attributes are necessary to demonstrate that a student has the technical skills required for employment when interviewing.
- The most significant negative attribute is "Backlogs" ($r = -0.303$). A student with a lot of backlogs is at the highest risk of not getting hired. Having a large number of backlogs can greatly reduce a student's chance of being hired, and often results in being automatically excluded from Corporate Hiring Cycles.

3. Confusion Matrix Inversion Analysis



- We created a classification confusion matrix in order to identify the prediction limits of the Stage 1 Placement Model on the independent validation data set ($n = 1800$). The matrix indicates a very high level of operational accuracy (confidently confirming employment-ready students). The model found 1466 True Positives (students who were successfully identified as Employable), and 136 True Negatives. The total margin of error was only 46 False Negatives and 152 False Positives. The above behaviour results in an overall classification accuracy of 89% for the tracking of employment. The statistical margin of safety reduces the chance of the web application generating False Positive placements for students who have difficulty or are not prepared to work.

4. Coordinate Density Mapping Interpretation



- The Volumetric Imbalance: In the picture the Vibrant Blue cloud that represents the 7,702 placed students is seen to be more crowded and tight when one looks at the Vibrant Yellow which is for the 1,298 unplaced students. We see that exactly 85.58% of the student population in this set has reported to have secure placements. Many think that the large size of the blue area is a proof of the human group's success. Data reports show that there are many more placed students which is what the greater blue area is a reflection of.
- Density Interpretation: While in all categories we use the same boundary which we have set to keep the layout clean and easy to scan through, we have made the Placed dots more transparent (0.2) which in turn allows you to see the large volume of what may be overlapping data points without it becoming a solid block. As for the Unplaced cloud we have used a higher transparency (0.6) which is required for its smaller set (14.42% to still be put forward and easily compared.

Conclusion

This research put forth and validated **PathFinder AI**, a hybrid multi-model machine learning platform which we designed to break down the skill-to-salary relationship and which also puts a stop to the issue of skills gap which computing graduates face in the Philippines. We used a raw set of **exactly 9,000** professional IT records for this, and the framework in turn changes the career assessment away from that of traditional academic credentials and toward more modern, skill-based metrics.

Through the introduction of a custom **Linear Inversion Normalization Layer**, which we put to use in the development of our framework, we were able to put into one scale what was very much diverse and different grading scales (which included 1.0–5.0 SUC metrics, private 4.0 scales, and percentage marks), transforming them into a standard $[0.0, 1.0]$ performance index which also in effect removed institutional evaluation bias.

We report that the results of this are very much transformative in terms of our study of the local IT hiring ecosystem:

- **The Primacy of Practical Skills:** From the algorithmic feature importance evaluations, we see that portfolio presentation (**resume_score at 23.18%**) and programming ability (**coding_score at 15.39%**) are identified as the primary factors in market hireability, playing a role that carries almost a fourfold (4x) higher predictive weight than traditional academic performance (**GWA at 9.67%**).
- **The Automated Barrier:** Academic shortcomings, represented by cumulative failed courses (**backlogs at 11.23%**), are shown to have a strong negative correlation of $r = -0.303$, playing the role of an automated screening factor which in turn greatly diminishes recruitment eligibility.
- **Environmental Impact on Salaries:** We saw from the Multiple Linear Regression analysis—which reported an **R^2 score of 0.7845** and a minimized **MAE of 0.0485**—that indeed corporate tiering plays a massive role in compensation scales. Putting a profile forward for a **Top Tech Titan firm sees a large positive premium ($\beta = +0.230$)** applied, whereas putting it forward for a

Bootstrap Startup role sees a severe financial discount ($\beta = -0.200$) for the exact same set of graduate skills.

- **Spatial Track Alignment:** We noted that the multi-class **K-Nearest Neighbors (K=5) engine** performed with high precision, achieving an overall classification accuracy of **85.94%** in routing student skill profiles into their optimal historical professional fit (**Software Engineer, Data Scientist, Web Developer, or Analyst**).

In the end, PathFinder AI has transformed university career counseling from a very subjective approach into a highly scientific and prescriptive simulation environment. The software we have developed establishes a scalable, objective, and data-driven platform which allows graduating computing professionals to identify their competency gaps, to fine-tune their project portfolios, and to strategically navigate the competitive compensation landscape of Industry 4.0.

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